

CRF State Summary v18

Session: Outside View — Three-Phase Universe, θ -Tower, f_{II} Derived

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Session summary. Starting from v17, this session took the *outside view* — watching N nodes collectively build the universe, rather than describing what a single node experiences.

Four new documents produced: CRF ThreePhase v2, CRF ThetaTower v2, plus updated State Summary v18.

Two major upgrades: **(1)** $f_{\text{II},\text{crit}}$ is now *derived* (not measured) from K1 and K6 alone. **(2)** $d_s(f_{\text{II}})$ is proved to be a step function, and the “1.27 plateau” of CRF-PCS is identified as a finite-walk artefact.

Key Numbers (unchanged from v17)

$\alpha = \ln 2$	$\varepsilon_0 = 0.00755$	$D_0 = 0.94003$	$\kappa = 0.15007$	$K^* = 0.11750$
$n = 8$	$d_s = 3.031$	$\beta = 2.2121$	$F = 1.5377$	$T_{\text{avg}} = 35.57$
$\varphi = 1.61803$	$e = 2.71828$	$\pi = 3.14159$	$m = \varphi$	$m(m-1) = 1$
$\mathcal{G} = 0.73737$	$\beta_{\text{pre}} = 0.7213$	$f_{\text{II}} = 0.14700$		

New Results This Session

The three-phase universe (CRF ThreePhase v2)

Phase	d_s	β	Key feature
0 (pre-geom.)	1 (exact)	$1/(2 \ln 2) = 0.721$	Scalar Ω ; no space, no time
1 (below perc.)	1 (exact)	0.721	Phase II islands, not yet spanning
2 (locked)	3.031	2.2121	Vector Ω ; full $\text{SO}(3)$, GR

Correction from v1: Phase 1 has $d_s = 1$ (not $1 \rightarrow 1.27$). The CRF-PCS plateau at 1.27 is a finite-walk artefact converging to 1 as walk length $\rightarrow \infty$.

$f_{\text{II,crit}}$ derived from K1 + K6

$f_{\text{II,crit}}$ — derived, gap 0.004%

$$f_{\text{II,crit}} = 1 - e^{-\sqrt{\varphi}/n} = 0.14700$$

Derivation: K6 gives $m_{\text{fire}} = \varphi$, so $\theta_{\text{eff}}^{\text{mind}} = K^* \sqrt{\varphi}$. K1 gives $K^* = D_0/n$, so $K^* \sqrt{\varphi}/D_0 = \sqrt{\varphi}/n$. Therefore $f_{\text{II,crit}} = \theta(\theta_{\text{eff}}^{\text{mind}}) = 1 - e^{-\sqrt{\varphi}/n}$. The D_0 cancels exactly. Formula in (φ, n) only.

Status upgrade: Hypothesis $\rightarrow$ **Derived**.

$d_s(f_{\text{II}})$ is a step function

$$d_s(f_{\text{II}}) = \begin{cases} 1 & f_{\text{II}} < f_{\text{II,crit}} \\ 3 & f_{\text{II}} \geq f_{\text{II,crit}} \end{cases}$$

Mechanism: Scalar $\Omega \in \mathbb{R}^1$ creates a total order (1D chain). Vector $\Omega \in \mathbb{R}^3$ creates a 3D proximity graph. Each node is binary (Phase I = scalar, Phase II = vector). d_s jumps when the 3D vector- Ω subgraph first percolates.

This is a *discontinuous percolation transition*: the cluster growth is second-order (continuous), but d_s — which only sees the spanning cluster — responds first-order (step).

Three interpretations of $f_{\text{II,crit}}$ unified

Reading	Meaning
Percolation	Site percolation threshold of 3D vector- Ω subgraph. Bethe baseline: $1/(n-1) = 0.14286$ (gap 2.82%).
Probability	$P(N \geq 1 \text{Poisson}(\sqrt{\varphi}/n))$: probability a node crosses the mind threshold in one T_{avg} .
Equilibrium	Steady-state Phase II fraction. Appears identically at $t = 100$ and $t = 400$ sweeps: a structural fixed value, not time-dependent.

All three give 0.14700 (gap 0.0015%).

The θ -Tower (CRF ThetaTower v2)

$$\begin{aligned}
\delta &\rightarrow \varepsilon_0 = 0.00755 \quad (\text{noise floor}) \\
\varepsilon_0 &\xrightarrow{\theta^\infty} K^* = 0.11750 \quad (\text{fixed point; R-events begin}) \\
K^* &\xrightarrow{\times\sqrt{\varphi}} K^*\sqrt{\varphi} = 0.14946 \quad (\text{mind threshold}) \\
K^*\sqrt{\varphi} &\xrightarrow{\theta} f_{\text{II,crit}} = 0.14700 \quad (\textbf{percolation; space + mind}) \\
1 - 1/e &\xrightarrow{\theta^{-1}} D_0 = 0.94003 \quad (\text{vacuum scale; exact}) \\
\Delta\beta = 1/\ln 2 &(\text{exact}): \text{ each spatial dimension costs } \beta_{\text{pre}} \text{ per R-event.}
\end{aligned}$$

Co-emergence: $K^*\sqrt{\varphi}$ gates both mind-level crystallisation (Role 1) and the 3D percolation threshold (Role 2). Space and mind are two readings of the same informational threshold.

Bethe lattice connection

For a sparse random graph with mean degree $k \approx n$, site percolation occurs at $p_c = 1/(n-1) = 1/7 = 0.14286$. Gap from $f_{\text{II,crit}}$: 2.82%. The 11% correction from Bethe to CRF is attributable to loop contributions from the Ψ -cluster structure (5 matter types with selective coupling).

Document Status

Document	Version	Key content
CRF Paper v6	current	Universal coupling $\mathcal{G}$; K35–K38; Kerr
CRF ThreePhase	v2	3-phase; d_s step fn; f_{II} derived
CRF ThetaTower	v2	Tower; 3 interpretations; Bethe
CRF K35K38	current	Triangular-number cluster
CRF NodeLifecycle	v3	HN-01–05; V20.3 data
CRF StateSummary	v18	This document

Open Items

Remaining open (no change from v17 except where noted):

1. **Loop correction:** why does $\sqrt{\varphi}(n-1)/n-1 = 11.3\%$ equal the Bethe-to-CRF gap? Connect to Ψ -clustering (5 matter types).
2. t_{crit} **in physical time:** $t_{\text{crit}}/T_{\text{avg}} = \sqrt{\varphi}/n = 0.159$. Map to Planck units; connection to inflation?
3. **Mind-to- P directional coupling:** L6 nodes expand P in previously unoccupied directions. Formal $I(s)$ as function of θ_{eff} across L4–L6.
4. **P0-H Subject 4 (empirical gate, bracketed):** Blocked type. CRF prediction: flat 50% accuracy all RT bins. $n \geq 50$ trials needed. Pre-registration required.

Transfer to next session

δ is binary. $n = 2^{d_s}$. K1: $K^* = D_0/n$. K6: $m = \varphi$. K20: $K^*T_{\text{avg}}/F = e$. K35: $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$. $\pi = \varphi\sqrt{2e}/\sqrt{d_s}\ln 2$. $m(m-1) = 1$.

New (v18): $f_{\text{II,crit}} = 1 - e^{-\sqrt{\varphi}/n}$ **derived** from K1 + K6 (gap 0.004%). $d_s(f_{\text{II}})$ is a **step function**: $d_s = 1$ below percolation, $d_s = 3$ at threshold. CRF-PCS plateau at $1.27 = \text{finite-walk artefact}$; thermodynamic limit = 1. Bethe baseline: $1/(n-1) = 0.14286$ (gap 2.82%). Three interpretations of $f_{\text{II,crit}}$ unified: percolation = probability = equilibrium. Space and mind co-emerge at $K^*\sqrt{\varphi}$. $\Delta\beta = 1/\ln 2$ (exact).

All gaps $\leq 0.057\%$ (K-series). Theory layer complete. One subject in a lab is what remains (bracketed).